

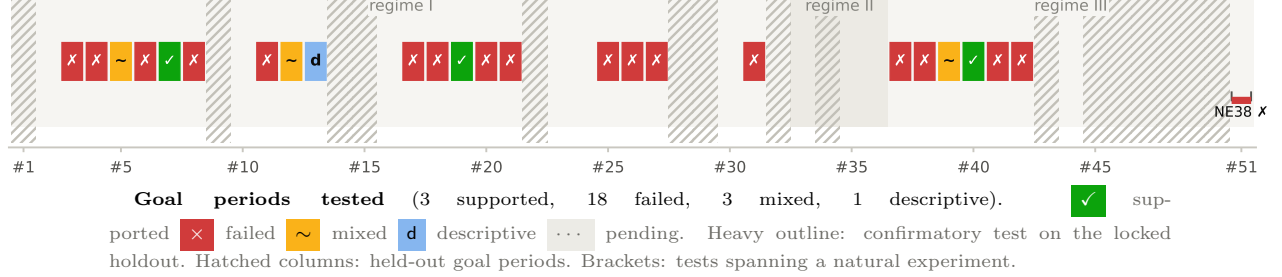
H96: Goal switches show hysteresis: a coercive field

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

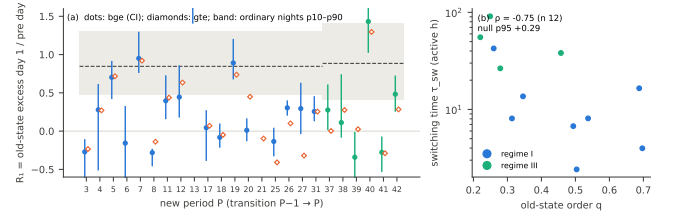
Bottom line. A goal switch quenches the swarm's content state within the first active hour, well beyond ordinary night-to-night drift; there is no sign of an order-dependent coercive field.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH123 claims that a goal switch shows hysteresis: the swarm keeps the old goal's state with a lag that grows with how ordered that state was, like a coercive field. H96 measures the old state's remanence after 24 non-holdout transitions (18 in regime I, 6 in regime III), with the new field projected out and placebo old states subtracted, in both embedding models. **A switch is a quench, not a hysteresis loop.** On day 1 the remanence is 0.27 (bge) / 0.18 (gte) of its pre-switch level, against 0.85–0.89 across an ordinary night (146 pseudo-switches). The order test is inconclusive: its synthetic power is 0.50, and the measured trend is negative ($\rho = -0.75$ / -0.52 , 12 transitions). One reassigned agent (NE38) loses its old role state faster than all 22 same-day placebos. The holdout script is written, not run.



Observables. (a) Persistence ratio R_1 per transition (bge with agent-bootstrap CI; gte diamonds). The grey band is the 10–90% range of ordinary nights. Most switches sit far below it. (b) Old-state order q against the switching time τ_{sw} for the 12 transitions with a measurable residual. HH123 predicts a rising trend; the data fall.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator who reassigns a swarm: there is nothing to clear first. The new kickoff removes 70–80% of the old state's content by the end of day 1, about 3× more than one night removes, and most of it within the first active hour. A single reassigned agent behaves the same way (NE38). A small residual of the old state can linger for about one active day (median τ_{sw} 7–15 active h), so expect stray references to the old goal on day 1, not a lag. The claim that ordered swarms resist a new goal is not supported. Its test is weak at this sample size, so it stays open, but the sign is wrong.

PHYSICAL MODEL

Model 11. Each agent is a unit spin $\mathbf{s}_i$ in the 32-d whitened, style-residualized content space. At the kickoff time t_0 the field steps from the old goal to the new one. The old state $\hat{e}_{old}$ is the leave-agent-out centroid of the old period's last days. $P_\perp$ removes the new field (kickoff, goal, room kickoffs). Placebo old states $\hat{e}_Q$ come from non-adjacent periods. Hysteresis predicts a decay time that grows with the old order q . The rivals are a quench and ordinary night-to-night drift.

$$M(t) = \langle \mathbf{s}_i(t) \cdot P_\perp \hat{e}_{old}^{(-i)} \rangle_i - \text{med}_Q \langle \mathbf{s}_i(t) \cdot P_\perp \hat{e}_Q^{(-i)} \rangle_i$$

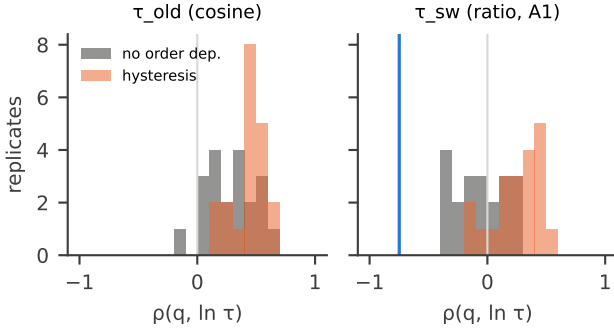
$$R_1 = M(\text{day } 1) / M(\text{pre}), \quad \Delta R_1 = R_1 - R_1^{\text{night}}$$

$$\ln \frac{M_{old}(h)}{M_{new}(h)} = L_0 - \frac{h}{\tau_{sw}}, \quad \text{HH123: } \ln \tau_{sw} = a + bq, \quad b > 0$$

H96-goal-switch-hysteresis · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H96-goal-switch-
hysteresis/README.md · ratings: H96 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed (bge / gte)	Verdict
P1 quench: $\Delta R_1 < 0$, -0.62 $[-0.78, -0.45]$ / -0.72		supported
R_1 below nights in $\geq [-0.88, -0.56]$; 20/23 / 21/24		
2/3		
P2' $\rho(q, \ln \tau_{sw}) > +0.29$	-0.75 / -0.52 (n 12); power 0.50	inconclusive
P3 median $\tau_{old} < 8$ ac-	0.16 / 0.15 h (cosine scale)	supported
tive h		
P4 old kickoff fades first	10/15 / 10/19	holds (weak)
N1 G39 room memory	ratio 0.15 / 0.26; ordered room keeps less	mixed
N2 NE38 single agent	R_1 0.36 / 0.22, percentile 0/22	quench
per transition	18 quench, 3 lag, 2 mixed, 1 desc. - (bge)	



Second figure. Synthetic transitions on the real skeletons (Amendment 1). Left: with the cosine decay time τ_{old} , $\rho(q, \ln \tau)$ is positive even without hysteresis, because unit normalization slows the cosine decay of large components. Right: the ratio time τ_{sw} is centred under the null. Its power at a doubling over the IQR is 0.50. The blue line is the real value.

GOAL PERIODS IN DETAIL

G03 × failed
G04 × failed
G05 ~ mixed
G06 × failed
G07 ✓ supported
G08 × failed
G11 × failed
G12 ~ mixed
G13 d descriptive
G17 × failed

+15 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The order test has power 0.50, so its negative is not a refutation, and its 12 transitions are selected (those with a measurable residual). The cosine decay time is biased by the old state's amplitude; only the ratio time is usable for cross-period comparison. The new field is a text-embedding proxy and operator messages are not removed. Composition carry-over is reduced, not removed. bge and gte disagree on some single transitions (#19→#20). Regime II has too few placebo old states. G39's room contrast is confounded by three agents who changed rooms. The holdout is not used.

ROUND 2

Fit a two-stage model (fast quench fraction plus slow residual) and ask whether the residual lives in veterans' carried-over context or in the record (newcomers, H82). Pool with H82's boundary rows to raise power.

- **H96-R1** Raise power for the order test: pool transitions with H82's boundary rows (same transitions, different estimator) as robustness, and add the held-out transitions in the confirm run (20 targets).
- **H96-R2** Two-stage model: fit a fast quench fraction plus a slow residual (two exponentials, or a ratio model with a quench offset) and test whether the residual is veterans' context carry-over (regime III) or record-borne (newcomers; H82 N1/N2).
- **H96-R3** Explain anti-coercivity: are ordered old states the ones whose new kickoff is most specific (H54's quench depth)?
- **H96-R4** Run `confirm.py` after Vivian's sign-off and the ledger disclosure (H82, H54, H10 plan the same targets).